



PES UNIVERSITY, BANGALORE
Department of Computer Science and Engineering

Software Requirements Specification

Hostel Management System

Version 1.0 approved

Prepared by Group 3

Name	SRN
Raina Kareem	PES1UG24CS361
Prema P Kotur	PES1UG24CS343
Praakruthi PS	PES1UG24CS324
Ojas Taori	PES1UG24CS308

PES University, Bangalore
Department of Computer Science and Engineering

04 October 2026



PES UNIVERSITY, BANGALORE
Department of Computer Science and Engineering

Table of Contents

1. Introduction	5
1.1 Purpose	5
1.2 Intended Audience and Reading Suggestions	5
1.3 Product Scope	5
1.4 Definitions, Acronyms and Abbreviations	6
1.5 References	6
1.6 Overview	7
2. Overall Description	8
2.1 Product Perspective	8
2.2 Product Functions	8
2.3 User Classes and Characteristics	9
2.4 Operating Environment	9
2.5 Design and Implementation Constraints	9
2.6 Assumptions and Dependencies	10
3. External Interface Requirements	11
3.1 User Interfaces	11
3.2 Software Interfaces	11
3.3 Communications Interfaces	12
3.4 Hardware Interfaces	12
3.5 Analysis Models	12
3.5.1 Use Case Diagram	12
3.5.2 Entity-Relationship Diagram	14
4. System Features (Functional Requirements)	15
4.1 User Authentication and Access Control	15



PES UNIVERSITY, BANGALORE
Department of Computer Science and Engineering

4.2 Student Profile Management	16
4.3 Hostel and Room Management	17
4.4 Room Application and Allocation	17
4.5 Room Change and Check-out	19
4.6 Complaints and Maintenance Requests	19
4.7 Fee Tracking	20
4.8 Reports and Dashboard	21
4.9 Notifications and Announcements	21
5. Other Nonfunctional Requirements	23
5.1 Performance Requirements	23
5.2 Safety Requirements	23
5.3 Security Requirements	23
5.3.1 Security Objectives	23
5.3.2 Security Requirements	24
5.4 Software Quality Attributes	24
5.5 Business Rules and Domain Requirements	25
6. Other Requirements	26
Appendix A: Glossary	27
Appendix B: Field Layouts	27
Appendix C: Requirement Traceability Matrix	31
Index	32



PES UNIVERSITY, BANGALORE
Department of Computer Science and Engineering

Revision History

Name	Date	Reason For Changes	Version
Group 3: Raina Kareem, Prema P Kotur, Praakruthi PS, Ojas Taori	04-10-2026	Initial SRS for Mini-Project Deliverables Part-1: introduction, overall description, interfaces, use case and ER models, functional, non-functional and security requirements, appendices.	1.0

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) describes the functional and non-functional requirements for release 1.0 of the **Hostel Management System (HMS)**, a web application that manages hostel room allocation and resident student details for a university campus. The document covers the complete HMS product; there are no sub-systems specified elsewhere. It follows the structure of IEEE Std 830-1998 as adapted in the PES University SRS template, and it is the baseline against which the Software Architecture and Design Specification and the Test Plan of this project will be traced.

1.2 Intended Audience and Reading Suggestions

The SRS is written for the following readers:

- **Course faculty and evaluators**, who assess completeness and quality of the requirements. Sections 4 and 5 hold the requirements; Section 3.5 holds the use case diagram; Section 5.3 holds the security objectives and requirements.
- **Developers (project team)**, who implement the system. Read Sections 2 and 3 for context and interfaces, then Section 4.
- **Testers**, who derive test cases. Every requirement carries a unique ID; Appendix C maps IDs to test cases.
- **Hostel administration (customer representatives)**: wardens and the hostel office. Sections 1.3, 2.2 and 5.5 are most relevant.
- **Maintainers**, who will support the system after delivery; see Sections 2.5, 5.4 and 6.

Section 1 introduces the product. Section 2 gives an overall description. Section 3 specifies external interfaces and analysis models. Section 4 lists functional requirements grouped by system feature. Section 5 lists non-functional requirements, including security, and the business and domain rules. Section 6 covers other requirements. Appendices A to C contain the glossary, field layouts and report requirements, and the requirement traceability matrix.

1.3 Product Scope

Many university hostels still allocate rooms using paper registers and spreadsheets. Students queue at the hostel office to apply, wardens track vacancies by hand, double allocations occur, and student contact details are scattered across files. HMS replaces this manual process with a single web application that lets students apply for rooms and maintain their details online, lets wardens allocate rooms against live vacancy data, and lets the administrator configure hostels, rooms and users.

The objectives of release 1.0 are to:

- reduce the time to process a room application from days to under 10 minutes of warden effort;
- eliminate double allocation of a bed through system-enforced constraints;
- keep one authoritative, searchable record of each resident's personal, guardian and emergency-contact details;

- give students transparent, self-service visibility of their application status, allocation, dues and complaints; and
- give hostel management accurate occupancy, fee-dues and complaint reports on demand.

Out of scope for release 1.0: online fee collection through a payment gateway (payments are recorded, not collected), mess/menu management, visitor and leave/out-pass management, biometric attendance, and a native mobile app (the web UI is responsive instead).

1.4 Definitions, Acronyms and Abbreviations

Term	Definition
HMS	Hostel Management System, the product specified in this document.
SRN	Student Registration Number, the unique university identifier of a student (e.g. PES1UG24CS361).
Warden	University staff member responsible for one or more hostels.
Allocation	Assignment of one student to one bed in one room for a period of time.
Bed	A single occupant slot within a room. A room of capacity N has N beds.
Application window	The date range, set by the Administrator, during which students may submit room applications.
Academic year	The period from 1 June to 31 May of the following year.
RBAC	Role-Based Access Control.
FR / NFR	Functional Requirement / Non-Functional Requirement.
CSRF, XSS	Cross-Site Request Forgery, Cross-Site Scripting (web attack classes).
OWASP	Open Worldwide Application Security Project.
DPDP Act	Digital Personal Data Protection Act, 2023 (India).
TBD	To Be Determined.

Further terms are defined in Appendix A.

1.5 References

- IEEE Std 830-1998, *IEEE Recommended Practice for Software Requirements Specifications*, IEEE, 1998.
- ISO/IEC/IEEE 29148:2018, *Systems and software engineering - Life cycle processes - Requirements engineering*.
- PES University, Dept. of CSE, *Software Requirements Specification template and SE Mini-Project Deliverables Part-1* (2026).
- Django Software Foundation, *Django 5.x Documentation*, <https://docs.djangoproject.com/>.
- OWASP Foundation, *OWASP Top 10:2021 and Application Security Verification Standard 4.0.3*, <https://owasp.org/>.
- W3C, *Web Content Accessibility Guidelines (WCAG) 2.1*, <https://www.w3.org/TR/WCAG21/>.
- Government of India, *Digital Personal Data Protection Act, 2023*.

1.6 Overview

The remainder of this document is organised as summarised in Section 1.2. Requirements are uniquely identified (for example FR-4.5, NFR-PERF-1, SEC-REQ-3) and written to be clear, unambiguous, concise and measurable so that each can be verified by a test.

2. Overall Description

2.1 Product Perspective

HMS is a new, self-contained product. It replaces the existing manual process (paper application forms, room registers and spreadsheets) used by the hostel office. The representative customer is a university hostel administration consisting of an administrator in the hostel office and wardens for each hostel block; its typical users are resident and prospective resident students.

HMS is a three-tier web application. Users access it through a web browser over HTTPS. The application tier (Django) contains the functional modules. Data is stored in a PostgreSQL database. Outgoing e-mail is sent through the university SMTP server. The student master list (SRN, name, gender, programme, year) is imported from the university records system as a CSV file; release 1.0 has no live integration with that system. Figure 1 shows the context.

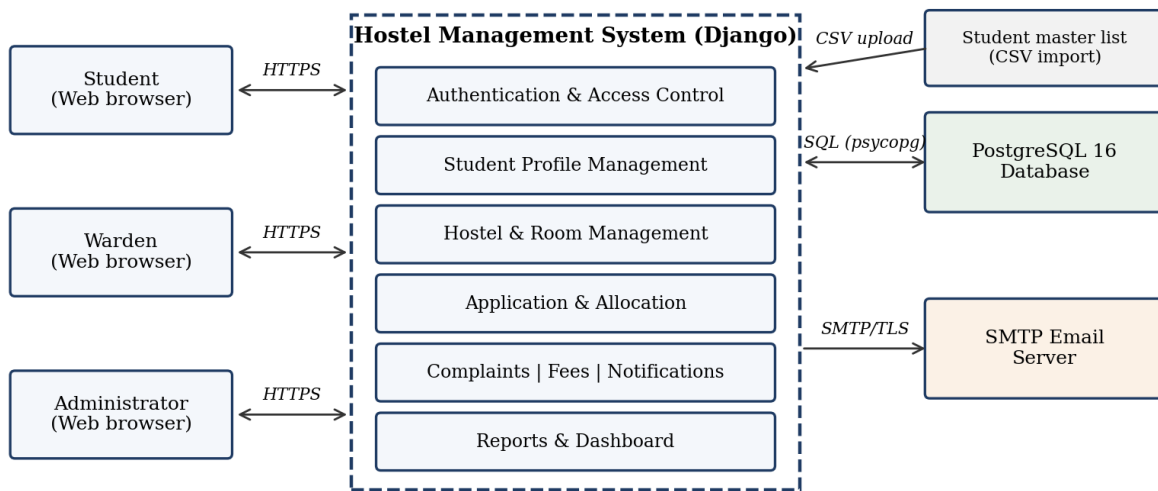


Figure 1: HMS context and major components

2.2 Product Functions

The major functions of HMS, detailed in Section 4, are:

- **Authentication and access control:** student self-registration against the master list, login, password reset, role-based access.
- **Student profile management:** maintain personal, academic, guardian and emergency-contact details; search residents.
- **Hostel and room management:** define hostels, floors, rooms, room types, capacity, fees and room status.
- **Room application and allocation:** students apply with preferences; wardens allocate beds manually or automatically.
- **Room change and check-out:** request and approve room changes; record vacating; keep allocation history.

- **Complaints:** raise and track maintenance complaints to resolution.
- **Fee tracking:** generate dues per allocation, record payments, issue receipts.
- **Reports and dashboard:** occupancy, residents list, fee dues and complaint reports.
- **Notifications and announcements:** e-mail on key events; hostel notices on the student dashboard.

2.3 User Classes and Characteristics

User class	Characteristics	Functions used	Importance
Student	About 2,000 users aged 17-25. Comfortable with web and mobile apps; low domain knowledge. Uses the system intensively during the application window and occasionally otherwise, often from a phone.	Register, profile, apply, view allocation and dues, room change, complaints, announcements.	Primary
Warden	About 10 staff. Moderate computer skills; strong domain knowledge. Uses the system daily from an office desktop.	Search students, review applications, allocate, room change, check-out, complaints, fees, reports, announcements.	Primary
Administrator	1-2 hostel-office staff with good computer skills. Highest privilege.	User accounts, hostels and rooms, application windows, master-list import, reports, audit log.	Secondary (few users, critical tasks)

2.4 Operating Environment

- **Server:** one virtual machine with 4 vCPU, 8 GB RAM, 100 GB SSD, running Ubuntu Server 24.04 LTS; Python 3.12; Django 5.x served by Gunicorn behind Nginx; PostgreSQL 16 on the same or a separate VM.
- **Client:** any device with a current version (latest two releases) of Google Chrome, Mozilla Firefox, Microsoft Edge or Apple Safari, at screen widths from 360 px (phones) to 1920 px (desktops). No client-side installation is required.
- **Network:** university campus network and the public Internet over HTTPS.

2.5 Design and Implementation Constraints

- C-1: The system shall be implemented in Python 3.12 using the Django 5.x framework, as specified in the course project list (Python Django/Flask or PHP Laravel).
- C-2: PostgreSQL 16 shall be the database; all access shall go through the Django ORM (no hand-built SQL strings).
- C-3: Only free and open-source libraries shall be used; there is no licence budget.
- C-4: The front end shall use server-rendered Django templates with Bootstrap 5; no separate single-page-application framework.
- C-5: Source code shall follow PEP 8 and be kept in a Git repository with all four team members committing.
- C-6: The system shall be deployable on a single university VM as described in Section 2.4.

- C-7: Handling of personal data shall comply with the DPDP Act, 2023.
- C-8: The project shall be delivered within one academic semester by a team of four students.

2.6 Assumptions and Dependencies

- A-1: The university provides a student master list (SRN, name, gender, programme, year, institutional e-mail) in CSV format before each application window.
- A-2: Every student has a working institutional e-mail address.
- A-3: A university SMTP relay is available for outgoing mail; if it is not, a third-party transactional e-mail service will be used.
- A-4: Fee payments are made through existing university channels; HMS only records them.
- A-5: Each hostel is designated for a single gender, and hostel rules (room types, fees, priority categories) are supplied by the hostel office before development of the related features.
- A-6: The university IT department provides the server VM, a domain name and a TLS certificate.
- D-1: The project depends on Django, psycopg, Bootstrap, ReportLab (PDF receipts and reports) and their continued availability.

3. External Interface Requirements

3.1 User Interfaces

- UI-1: Every page shall show a top navigation bar with the HMS name, the menu items allowed for the logged-in role, the user name, and Logout; and a footer with a Help link and contact details of the hostel office.
- UI-2: After login each role lands on its own dashboard: Student (application status, allocation, dues, open complaints, announcements); Warden (vacancies, pending applications, complaints older than 72 hours); Administrator (system summary).
- UI-3: Forms shall mark mandatory fields with an asterisk, validate on submission, show each error in red directly beneath the offending field, and list a summary of errors at the top of the form. Valid input already entered shall be preserved.
- UI-4: Destructive or irreversible actions (reject, check-out, deactivate) shall require confirmation in a modal dialog.
- UI-5: Success and failure of every action shall be reported in a dismissible banner message.
- UI-6: Tables with more than 25 rows shall be paginated (25 rows per page) and sortable by column.
- UI-7: The layout shall be responsive and usable without horizontal scrolling at 360 px width.
- UI-8: Dates shall be shown as DD-MM-YYYY and amounts as INR with two decimal places.

Principal screens: Login, Register, Forgot/Reset Password, Dashboards (3), My Profile, Apply for Room, My Allocation and Dues, Room Change Request, Complaints, Announcements, Student Search, Applications Queue, Room Availability Grid, Allocate Room, Record Payment, Reports, Manage Hostels and Rooms, Manage Users, Application Windows, Import Master List, Audit Log.

3.2 Software Interfaces

Component	Version	Purpose / data exchanged
Django	5.x	Web framework: routing, ORM, authentication, templates, CSRF protection.
PostgreSQL	16	Persistent store for all entities (Section 3.5.2). Accessed via psycopg3 over TCP port 5432 (local network only).
SMTP server	University relay	Outgoing notification e-mails (Section 4.9). HMS sends; it never receives mail.
ReportLab	4.x	Generation of PDF fee receipts and reports.
Bootstrap	5.3	Responsive CSS and UI components (served locally, not from a CDN).
Gunicorn / Nginx	22.x / 1.24+	WSGI application server / reverse proxy and TLS termination.
Ubuntu Server	24.04 LTS	Server operating system.
University records system	N/A	Source of the student master list, received as a CSV file uploaded by the Administrator.

Internal JSON endpoints (used by the browser for the room availability grid and live search) are described in the Software Architecture and Design Specification. All shared data resides in the

database; there is no shared memory between processes.

3.3 Communications Interfaces

- COM-1: All browser-server communication shall use HTTPS (TLS 1.2 or higher) on port 443; requests to port 80 shall be redirected to HTTPS.
- COM-2: Asynchronous browser requests shall exchange UTF-8 encoded JSON over the same HTTPS channel.
- COM-3: E-mail shall be sent over SMTP with STARTTLS on port 587, in plain text and HTML, UTF-8 encoded.
- COM-4: The database connection shall not be exposed outside the server's private network.

3.4 Hardware Interfaces

HMS has no direct hardware interfaces. It requires only the standard server and client hardware listed in Section 2.4. Printing of receipts and reports is done through the browser's print function from generated PDF files.

3.5 Analysis Models

3.5.1 Use Case Diagram

Figure 2 shows the actors and use cases of HMS. All human actors authenticate through *Login / Logout* before using any other use case. *Apply for Room* and *Allocate Room* both include *Check Room Availability*; *Allocate Room* includes *Send Notification*, which uses the external Email Service.

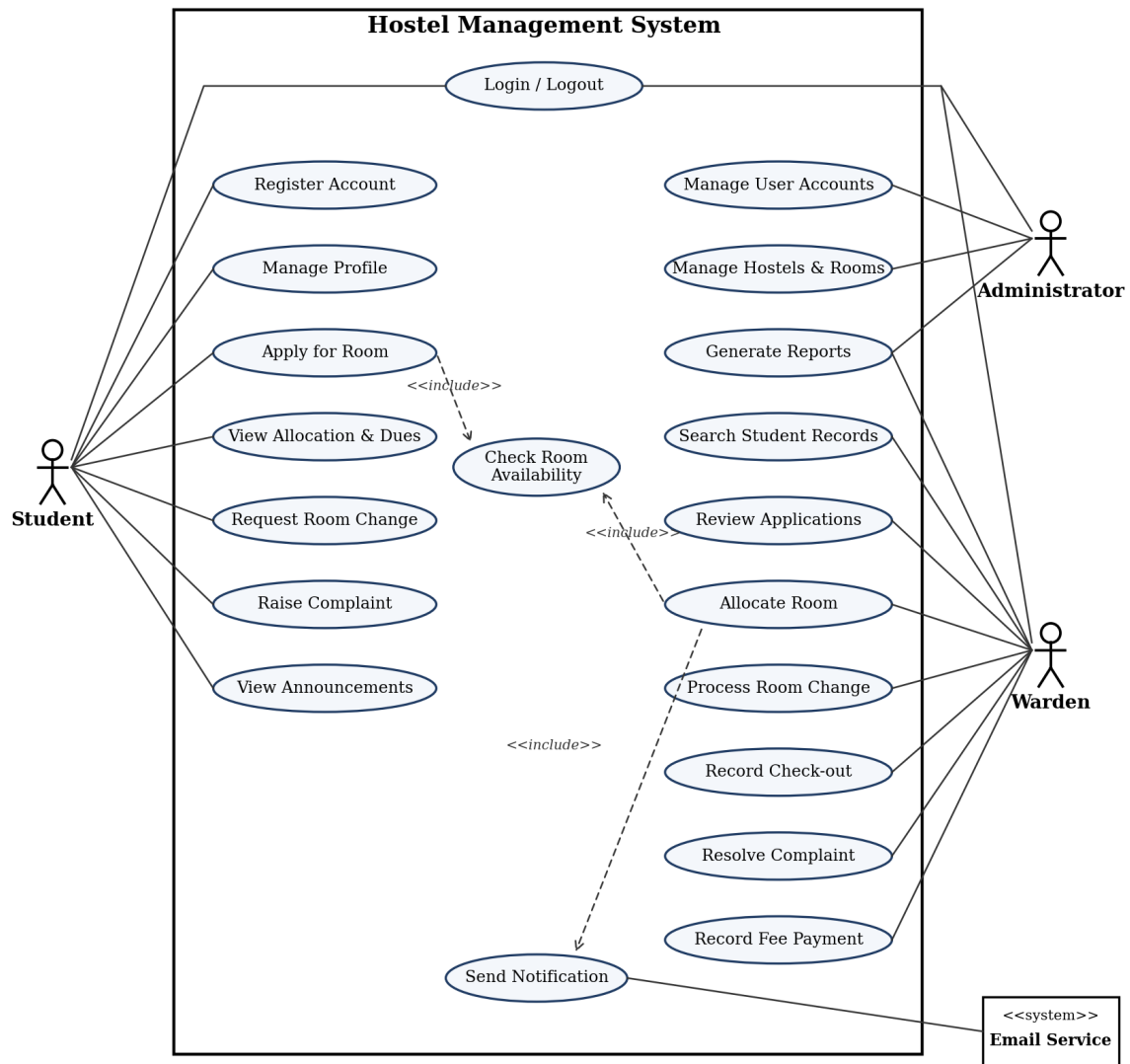


Figure 2: UML use case diagram of the Hostel Management System

Use case	Primary actor	Brief description	Requirements
Apply for Room	Student	Student submits an application with up to three room-type preferences during the open window.	FR-4.1 - FR-4.3
Allocate Room	Warden	Warden assigns a vacant bed in a gender-matching hostel to an approved applicant; student is notified.	FR-4.4 - FR-4.8
Request / Process Room Change	Student / Warden	Student requests a move; warden approves and the system moves the allocation atomically.	FR-5.1 - FR-5.3
Raise / Resolve Complaint	Student / Warden	Student logs a maintenance issue; warden updates it to resolution.	FR-6.1 - FR-6.5
Manage Hostels & Rooms	Administrator	Create and maintain hostels, rooms, capacities and fees.	FR-3.1 - FR-3.6

3.5.2 Entity-Relationship Diagram

Figure 3 shows the main entities and relationships. Field details are given in Appendix B.

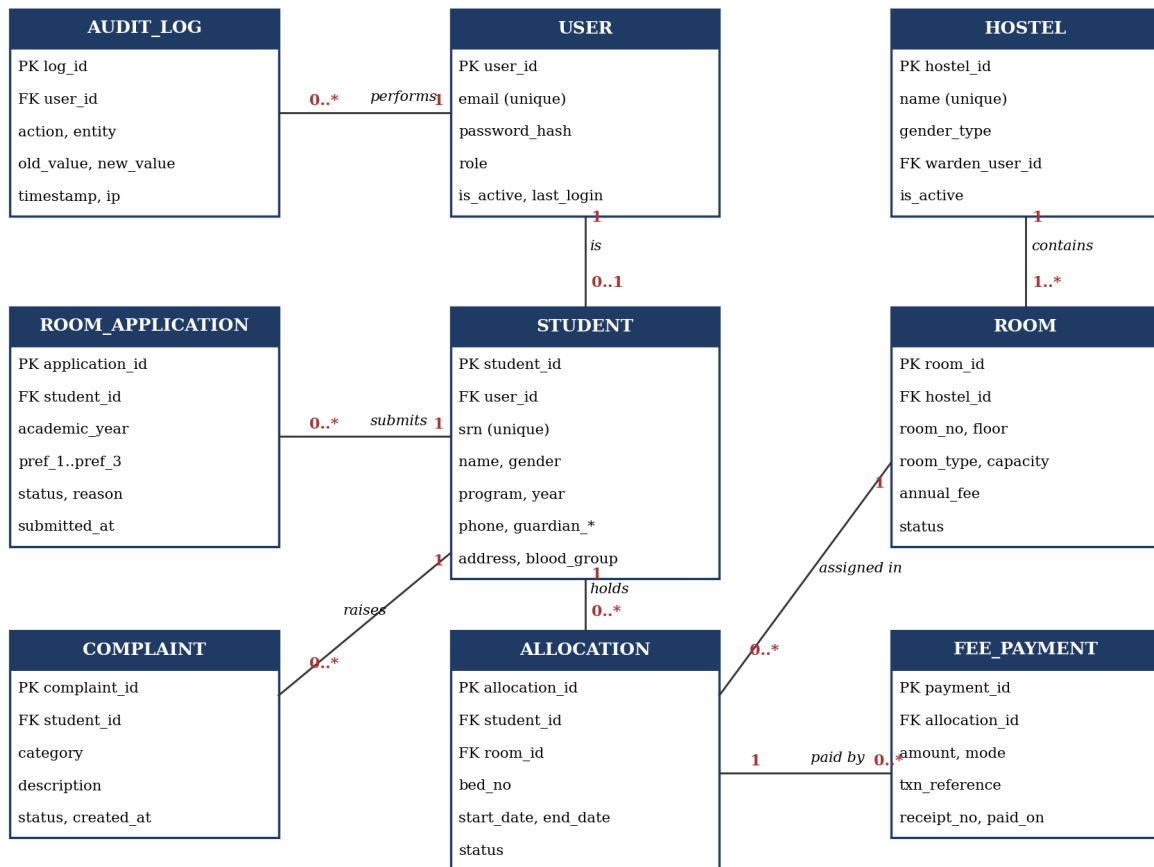


Figure 3: ER diagram of the HMS database

4. System Features (Functional Requirements)

Functional requirements are organised by system feature. "Shall" denotes a mandatory requirement. Each requirement is uniquely identified as FR-<feature>.<number> and is traced to design and test artefacts in Appendix C.

4.1 User Authentication and Access Control

4.1.1 Description and Priority

Allows students to create accounts and all users to log in securely, with access limited to the functions of their role. **Priority: High** (Benefit 9, Penalty 9, Cost 4, Risk 3; scale 1-9).

4.1.2 Stimulus/Response Sequences

#	Stimulus (user action)	Response (system)
1	Student opens Register and enters SRN, institutional e-mail and password.	Verifies SRN and e-mail against the master list, creates the account, sends a verification e-mail.
2	User enters e-mail and password on Login.	Authenticates and redirects to the role dashboard, or shows "Invalid e-mail or password".
3	User clicks Forgot Password and enters e-mail.	Sends a single-use reset link if the account exists; shows the same message either way.
4	User is idle for 30 minutes.	Ends the session and redirects to Login on the next request.

4.1.3 Functional Requirements

ID	Requirement
FR-1.1	The system shall allow a student to self-register only when the SRN and institutional e-mail entered match a record in the imported student master list and no account exists for that SRN; otherwise it shall show "SRN not found or already registered".
FR-1.2	The system shall activate a new student account only after the student opens the verification link e-mailed to the institutional address; the link shall expire after 24 hours.
FR-1.3	The system shall authenticate users by e-mail address and password and shall show the same generic error message for an unknown e-mail and a wrong password.
FR-1.4	The system shall assign each account exactly one role (Student, Warden or Administrator) and shall allow access to each page and endpoint only for the roles listed in the permission matrix of BR-1 (Section 5.5).
FR-1.5	The system shall lock an account for 15 minutes after 5 consecutive failed login attempts made within 10 minutes and shall inform the user that the account is temporarily locked.
FR-1.6	The system shall provide password reset through a single-use link, sent to the registered e-mail, that expires 30 minutes after issue.
FR-1.7	The system shall end a session after 30 minutes of inactivity and immediately on Logout, invalidating the session on the server.
FR-1.8	Only the Administrator shall be able to create Warden and Administrator accounts, change roles, and deactivate any account.

4.2 Student Profile Management

4.2.1 Description and Priority

Maintains one authoritative record of each student's personal, academic, guardian and emergency-contact details and lets staff find residents quickly. **Priority: High** (Benefit 8, Penalty 8, Cost 3, Risk 2; scale 1-9).

4.2.2 Stimulus/Response Sequences

#	Stimulus (user action)	Response (system)
1	Student opens My Profile, edits phone number and guardian details, clicks Save.	Validates fields, saves, records the change in the audit log, shows "Profile updated".
2	Student enters an invalid phone number.	Rejects the save and shows the field-level error "Enter a valid 10-digit mobile number".
3	Warden types part of a name or an SRN in Student Search.	Shows matching students with hostel, room and contact numbers.

4.2.3 Functional Requirements

ID	Requirement
FR-2.1	The system shall allow a student to view all of his/her profile fields and to edit: mobile number, personal e-mail, permanent address, guardian name, guardian relationship, guardian mobile number, emergency contact name and number, blood group, and photograph.
FR-2.2	SRN, name, gender, programme and year of study shall be read-only for students and editable only by the Administrator.
FR-2.3	The system shall validate profile input as follows: mobile numbers are exactly 10 digits beginning with 6, 7, 8 or 9; e-mail addresses are syntactically valid; address is 10-250 characters; photograph is a JPEG or PNG of at most 2 MB. Invalid input shall be rejected with a field-level message and nothing shall be saved.
FR-2.4	The system shall require all mandatory profile fields (Appendix B, Table B-1) to be complete before a student can submit a room application.
FR-2.5	The system shall let Wardens and the Administrator search students by full or partial SRN, full or partial name (case-insensitive), hostel, room number, or year of study, and shall show SRN, name, hostel, room, mobile and guardian mobile in the results.
FR-2.6	A Warden shall see only students who have applied to or are allocated in the hostels assigned to that Warden.
FR-2.7	The system shall record every profile change in the audit log with field name, old value, new value, user and timestamp.

4.3 Hostel and Room Management

4.3.1 Description and Priority

Lets the Administrator define the physical inventory (hostels, rooms, beds) and keeps occupancy status accurate automatically. **Priority: High** (Benefit 9, Penalty 9, Cost 3, Risk 2; scale 1-9).

4.3.2 Stimulus/Response Sequences

#	Stimulus (user action)	Response (system)
1	Administrator adds a room with number, floor, type, capacity and fee.	Validates uniqueness and ranges, creates the room with status Available.
2	Last vacant bed of a room is allocated.	Sets room status to Full automatically.
3	Administrator tries to deactivate a room with occupants.	Blocks the action with "Room has active allocations".

4.3.3 Functional Requirements

ID	Requirement
FR-3.1	The system shall allow the Administrator to create, edit and deactivate hostels, each with a unique name, a gender designation (Male or Female), and one or more assigned Wardens.
FR-3.2	The system shall allow the Administrator to create and edit rooms with: room number (unique within its hostel), floor (0-20), room type (Single, Double or Triple), capacity (1-4), annual fee (INR 0-5,00,000) and status (Available, Full or Under Maintenance).
FR-3.3	The system shall compute room occupancy as the number of active allocations in the room, shall set status to Full when occupancy equals capacity, and to Available when a bed is released, unless the room is Under Maintenance.
FR-3.4	The system shall not allocate beds in rooms whose status is Under Maintenance.
FR-3.5	The system shall prevent deleting or deactivating a hostel or room that has active allocations, and shall never physically delete hostel or room records.
FR-3.6	The system shall show a room availability grid filterable by hostel, floor and room type, displaying capacity, occupancy and vacant beds for each room.
FR-3.7	The system shall allow the Administrator to import rooms from a CSV file, validating every row against FR-3.2 and producing a report of accepted and rejected rows with the reason for each rejection.

4.4 Room Application and Allocation

4.4.1 Description and Priority

The core feature: students apply for accommodation and wardens allocate beds, without double booking and with prompt notification. **Priority: High** (Benefit 9, Penalty 9, Cost 6, Risk 6; scale 1-9).

4.4.2 Stimulus/Response Sequences

#	Stimulus (user action)	Response (system)
1	Administrator sets an application window for the academic year.	Opens the Apply for Room screen to eligible students for that period.

#	Stimulus (user action)	Response (system)
2	Student submits an application with three room-type preferences.	Saves it as Pending, shows the application number and acknowledges by e-mail.
3	Warden opens the Applications Queue and selects an application.	Shows student details and only beds that are vacant in gender-matching hostels.
4	Warden selects a bed and clicks Allocate.	Creates the allocation as Provisional, updates occupancy, creates the fee record, notifies the student.
5	Two wardens try to allocate the same bed at the same moment.	Commits the first request and rejects the second with "Bed no longer available".

4.4.3 Functional Requirements

ID	Requirement
FR-4.1	The system shall allow the Administrator to define one application window per academic year (start date and end date) and shall accept student applications only within that window.
FR-4.2	The system shall allow a student with a complete profile to submit one application per academic year containing up to three distinct room-type preferences in order of priority and an optional preferred roommate SRN.
FR-4.3	The system shall reject a second application by the same student for the same academic year, and any application outside the window, with an explanatory message.
FR-4.4	The system shall list pending applications to Wardens ordered by priority category (BR-4) and then by submission time, oldest first, and filterable by year of study and room-type preference.
FR-4.5	When a Warden allocates, the system shall offer only beds that are vacant, in rooms that are not Under Maintenance, in hostels whose gender designation matches the student's gender.
FR-4.6	The system shall guarantee that a bed is never held by more than one active allocation and that a student never has more than one active allocation, including under concurrent requests, by enforcing these rules with database constraints inside a single transaction.
FR-4.7	The system shall provide an Auto-Allocate action that processes all pending applications for a hostel in the order of FR-4.4, assigns each student a vacant bed of the highest available preference, and displays a summary of allocated and unallocated applications.
FR-4.8	A Warden shall be able to reject an application only after entering a reason of at least 10 characters.
FR-4.9	The system shall create each new allocation with status Provisional, change it to Confirmed when the fee for the allocation is fully recorded as paid (FR-7.3), and show the current status to the student.
FR-4.10	The system shall e-mail the student within 5 minutes of an application being allocated or rejected (FR-9.1).

4.5 Room Change and Check-out

4.5.1 Description and Priority

Supports movement of residents between rooms and the vacating of rooms while keeping a full history.

Priority: Medium (Benefit 6, Penalty 5, Cost 4, Risk 4; scale 1-9).

4.5.2 Stimulus/Response Sequences

#	Stimulus (user action)	Response (system)
1	Student submits a room change request with a reason.	Saves it as Pending if no other request is pending and BR-6 is satisfied.
2	Warden approves the request and chooses a new bed.	Ends the old allocation and creates the new one in one transaction; notifies the student.
3	Warden records check-out for a student.	Ends the allocation with the check-out date and frees the bed immediately.

4.5.3 Functional Requirements

ID	Requirement
FR-5.1	The system shall allow a student with a Confirmed allocation to submit a room change request with a reason of 10-300 characters, with at most one pending request per student.
FR-5.2	On approval, the system shall end the current allocation and create the new allocation in a single atomic transaction, so that either both changes are saved or neither is.
FR-5.3	A Warden shall be able to reject a room change request with a reason, which shall be shown to the student.
FR-5.4	The system shall allow a Warden to record a check-out with a date not later than today, ending the allocation and releasing the bed immediately.
FR-5.5	The system shall retain every past allocation (room, bed, start date, end date, reason for ending) and show the history to the student and to Wardens.

4.6 Complaints and Maintenance Requests

4.6.1 Description and Priority

Lets residents report room or facility problems and lets wardens track them to closure. **Priority: Medium** (Benefit 6, Penalty 4, Cost 3, Risk 2; scale 1-9).

4.6.2 Stimulus/Response Sequences

#	Stimulus (user action)	Response (system)
1	Student raises a complaint with a category, description and optional photo.	Creates a ticket with a unique ID and status Open; shows it in the student's list.
2	Warden changes ticket status to Resolved with a remark.	Saves the change, notifies the student by e-mail.

4.6.3 Functional Requirements

ID	Requirement
FR-6.1	The system shall allow a student with an active allocation to raise a complaint with a category (Electrical, Plumbing, Furniture, Cleanliness, Internet or Other), a description of 20-500 characters, and an optional JPEG/PNG photograph of at most 2 MB.
FR-6.2	The system shall assign each complaint a unique ID of the form CMP-YYYY-NNNNN and record its hostel and room automatically from the allocation.
FR-6.3	A Warden shall be able to change a complaint's status only along Open to In Progress to Resolved, or Open/In Progress to Rejected, entering a remark with each change.
FR-6.4	The student shall be able to reopen a Resolved complaint within 7 days of resolution.
FR-6.5	The system shall highlight on the Warden dashboard every complaint that has been Open or In Progress for more than 72 hours.

4.7 Fee Tracking

4.7.1 Description and Priority

Records hostel fee dues and payments against allocations. Collection itself happens outside HMS (assumption A-4). **Priority: Medium** (Benefit 7, Penalty 6, Cost 3, Risk 3; scale 1-9).

4.7.2 Stimulus/Response Sequences

#	Stimulus (user action)	Response (system)
1	Allocation is created.	Creates a fee record for the academic year equal to the room's annual fee.
2	Warden records a payment with amount, mode and reference.	Validates the amount, updates the balance, generates a numbered PDF receipt.
3	Student opens My Allocation and Dues.	Shows total fee, amount paid, balance and payment history with receipts.

4.7.3 Functional Requirements

ID	Requirement
FR-7.1	The system shall create a fee record for each new allocation equal to the annual fee of the allocated room for that academic year.
FR-7.2	The system shall allow a Warden or the Administrator to record a payment with amount (greater than 0), mode (Online transfer, Demand draft or Cash), transaction reference (mandatory unless mode is Cash) and payment date.
FR-7.3	The system shall reject a payment that exceeds the outstanding balance and, when the balance becomes zero, shall mark the fee record as Paid and the allocation as Confirmed.
FR-7.4	The system shall generate a PDF receipt with a unique sequential receipt number for every recorded payment, downloadable by the student and staff.
FR-7.5	The system shall show each student the total fee, amount paid, balance and payment history for every allocation.
FR-7.6	The system shall list for Wardens all Provisional allocations for which no payment has been recorded 15 days after allocation (BR-5).

4.8 Reports and Dashboard

4.8.1 Description and Priority

Gives hostel management an accurate operational picture. **Priority: Medium** (Benefit 6, Penalty 4, Cost 3, Risk 2; scale 1-9).

4.8.2 Stimulus/Response Sequences

#	Stimulus (user action)	Response (system)
1	Warden opens the dashboard.	Shows total beds, occupied beds, vacant beds, occupancy %, pending applications and overdue complaints for the Warden's hostels.
2	Administrator requests the Fee Dues report for a hostel and exports it to CSV.	Generates the report and downloads the CSV file.

4.8.3 Functional Requirements

ID	Requirement
FR-8.1	The system shall display role-specific dashboards with the figures listed in UI-2, computed from current data at the time the page loads.
FR-8.2	The system shall generate the following reports, filterable by hostel and date range: Occupancy, Resident List, Fee Dues and Complaint Summary (fields in Appendix B, Table B-6).
FR-8.3	The system shall export every report to CSV and PDF.
FR-8.4	Wardens shall be able to generate reports only for their assigned hostels; the Administrator for all hostels.

4.9 Notifications and Announcements

4.9.1 Description and Priority

Keeps students informed without requiring them to check the system constantly. **Priority: Low** (Benefit 5, Penalty 3, Cost 2, Risk 2; scale 1-9).

4.9.2 Stimulus/Response Sequences

#	Stimulus (user action)	Response (system)
1	An application is allocated or rejected.	Queues an e-mail to the student.
2	Warden posts an announcement with an expiry date.	Shows it on the dashboard of every resident of that hostel until the expiry date.

4.9.3 Functional Requirements

ID	Requirement
FR-9.1	The system shall send an e-mail to the affected user for: account verification, password reset, application received, application allocated or rejected, room change approved or rejected, and complaint status change.
FR-9.2	The system shall retry a failed e-mail up to 3 times at 5-minute intervals and record each final failure in the system log.

ID	Requirement
FR-9.3	A Warden shall be able to post announcements (title up to 100 characters, body up to 2,000 characters, expiry date) to residents of the Warden's hostels; the system shall stop displaying them after the expiry date.

5. Other Nonfunctional Requirements

Unless stated otherwise, measurements are taken on the reference server of Section 2.4 with a database of 2,000 students, 1,500 beds and one year of transaction data.

5.1 Performance Requirements

ID	Requirement
NFR-PERF-1	With 200 concurrent users, 95% of page requests shall complete within 2 seconds and 100% within 5 seconds, measured at the server (load test with Locust).
NFR-PERF-2	The allocation transaction (FR-4.6) shall complete within 1 second for 95% of requests.
NFR-PERF-3	Student search (FR-2.5) shall return results within 2 seconds for a database of 5,000 student records.
NFR-PERF-4	Auto-Allocate (FR-4.7) shall process 500 pending applications within 60 seconds.
NFR-PERF-5	Any report of up to 5,000 rows (FR-8.2) shall be generated and exported within 10 seconds.

Rationale: demand peaks on the opening day of the application window, when most of the ~2,000 students log in within a few hours.

5.2 Safety Requirements

ID	Requirement
NFR-SAFE-1	The database shall be backed up automatically every day at 02:00 IST; backups shall be retained for 30 days and stored on a different machine. A restore shall be tested at least once a month. (Recovery point objective 24 hours; recovery time objective 4 hours.)
NFR-SAFE-2	The system shall never allocate a student to a hostel whose gender designation differs from the student's gender (BR-2), including through Auto-Allocate and room change.
NFR-SAFE-3	Records shall be soft-deleted only (marked inactive) so that accidental deactivation can be reversed by the Administrator.
NFR-SAFE-4	A Warden shall be able to obtain a printable list of all residents of a hostel with room number, mobile number, guardian mobile number and blood group in at most 3 clicks from the dashboard, for use in emergencies.

5.3 Security Requirements

HMS stores personal data of students (contact details, home addresses, guardian information, fee status). The security objectives below state what must be protected; the security requirements state how each objective is met verifiably. Security validation is planned in Section 5.1 of the Test Plan.

5.3.1 Security Objectives

ID	Objective	Description
SEC-OBJ-1	Confidentiality	Personal and fee data of a student shall be disclosed only to that student and to authorised staff of the student's hostel and the hostel office.
SEC-OBJ-2	Integrity	Allocation, fee and profile records shall be changed only by authorised users through defined functions, and unauthorised changes shall be prevented or detected.

ID	Objective	Description
SEC-OBJ-3	Authentication	Only verified users shall gain access, and accounts shall be resistant to password guessing and session hijacking.
SEC-OBJ-4	Accountability	Every security-relevant and privileged action shall be attributable to an individual user.
SEC-OBJ-5	Availability	The service shall remain available during the application window despite failures and common attacks.

5.3.2 Security Requirements

ID	Requirement	Objective
SEC-REQ-1	Passwords shall be stored only as salted hashes using PBKDF2-SHA256 with at least 600,000 iterations or Argon2id; plaintext or reversibly encrypted passwords shall never be stored or logged.	OBJ-1, 3
SEC-REQ-2	Passwords shall be at least 10 characters, contain at least one letter and one digit, differ from the user's e-mail and SRN, and not appear in a list of the 20,000 most common passwords.	OBJ-3
SEC-REQ-3	All traffic shall use HTTPS with TLS 1.2 or higher; HTTP Strict Transport Security shall be enabled with max-age of at least 31,536,000 seconds.	OBJ-1, 3
SEC-REQ-4	Authorisation shall be checked on the server for every request according to BR-1. A request for a resource outside the user's role or hostel scope shall return HTTP 403 (or 404 for object-level access) and be logged. Changing an ID in a URL shall never expose another student's data.	OBJ-1, 2
SEC-REQ-5	Session cookies shall be flagged Secure, HttpOnly and SameSite=Lax; the session identifier shall be regenerated at login; sessions shall expire as in FR-1.7.	OBJ-3
SEC-REQ-6	Every state-changing request shall be protected by a CSRF token; all output shall be HTML-escaped; all database access shall use parameterised ORM queries. The release build shall have zero High or Critical findings in an OWASP ZAP baseline scan.	OBJ-2
SEC-REQ-7	Repeated login failures shall trigger account lock-out as in FR-1.5, and password-reset requests shall be limited to 5 per e-mail address per hour.	OBJ-3, 5
SEC-REQ-8	The system shall write an append-only audit log entry for: login success and failure, logout, password change, role change, allocation create/change/end, payment recorded, profile change and record deactivation. Entries shall contain user, action, entity, old and new values, timestamp and IP address, be retained for 1 year, and be viewable only by the Administrator.	OBJ-2, 4
SEC-REQ-9	Uploaded files shall be accepted only if their content (magic bytes) is JPEG, PNG or PDF and their size is at most 2 MB; they shall be stored outside the web root under random file names and served only to authorised users.	OBJ-1, 2
SEC-REQ-10	Database credentials, the Django secret key and SMTP passwords shall be supplied through environment variables and shall not appear in the source code repository.	OBJ-1
SEC-REQ-11	Only fields listed in Appendix B shall be collected (data minimisation, DPDP Act 2023); a student shall be able to download his/her own profile and allocation data as a PDF.	OBJ-1

5.4 Software Quality Attributes

ID	Requirement
NFR-USE-1	Usability: in a test with at least 10 first-time student users, at least 80% shall complete a room application without assistance within 5 minutes of logging in.
NFR-USE-2	Accessibility: all pages shall meet WCAG 2.1 Level AA for colour contrast, text alternatives for images and full keyboard operability.
NFR-AVL-1	Availability: the system shall be available at least 99% of the time between 07:00 and 23:00 IST each month, and at least 99.5% during the application window, excluding announced maintenance.
NFR-REL-1	Reliability: the nightly consistency check shall find zero beds with more than one active allocation and zero rooms whose stored status disagrees with computed occupancy.
NFR-MNT-1	Maintainability: code shall pass flake8 with zero errors; automated unit tests shall cover at least 70% of statements in the allocation, fee and access-control modules.
NFR-PORT-1	Portability: the system shall function identically on the browsers listed in Section 2.4 and at screen widths from 360 px to 1920 px.
NFR-SCL-1	Scalability: the system shall support up to 5,000 students and 4,000 beds with no change to code, meeting NFR-PERF-1 on a server with twice the reference resources.

Where attributes conflict, correctness of allocation data takes precedence over performance, and ease of use for students takes precedence over ease of learning for staff.

5.5 Business Rules and Domain Requirements

BR-1 Permission matrix. Only the roles marked may perform each function:

Function	Student	Warden	Admin
Register account; edit own profile; apply for room; request room change; raise complaint	Yes	No	No
View own allocation, dues and receipts	Yes	No	No
Search and view student records	No	Own hostels	All
Review applications; allocate; approve room change; record check-out	No	Own hostels	All
Resolve complaints; post announcements	No	Own hostels	All
Record fee payments	No	Own hostels	All
Generate reports	No	Own hostels	All
Manage hostels, rooms, application windows, master list, user accounts, roles	No	No	Yes
View audit log; edit SRN, name, gender, programme, year	No	No	Yes

- **BR-2** A student may be allocated only to a hostel whose gender designation matches the student's gender as recorded in the master list.
- **BR-3** A student may hold at most one active allocation and one application per academic year.
- **BR-4** Allocation priority (domain rule from the hostel office): first-year students first, then students with an approved medical/disability priority flag set by the Administrator, then all others; within each category, first come first served by submission time.

- **BR-5** An allocation remains Provisional until the annual fee is paid in full. A Warden may cancel a Provisional allocation with no payment recorded 15 days after allocation; the bed then becomes available.
- **BR-6** A room change request may not be submitted within the first 30 days of an allocation, except when the Warden initiates the move.
- **BR-7** Check-out may be recorded only when the fee balance is zero, unless the Warden enters an override reason, which is logged.
- **BR-8** The academic year runs from 1 June to 31 May; all allocations end on 31 May unless renewed through a new application.
- **BR-9** Room capacities and fees are set only by the Administrator, in line with the hostel office's approved fee schedule.

6. Other Requirements

- **OR-1 Database:** The schema shall be in third normal form, use UTF-8 encoding, and enforce referential integrity with foreign keys and the uniqueness rules of FR-4.6 through database constraints. All timestamps shall be stored in UTC and displayed in IST.
- **OR-2 Data retention:** Records of students who have left shall be retained for 3 years after their last allocation ends and then anonymised.
- **OR-3 Legal:** The system shall display a privacy notice at registration describing what personal data is collected and why, as required by the DPDP Act, 2023, and record the student's acknowledgement.
- **OR-4 Internationalisation:** Release 1.0 shall be in English only, but all user-visible strings shall be marked for translation using Django's i18n framework.
- **OR-5 Documentation:** The team shall deliver a user guide for each role and a deployment guide.

Appendix A: Glossary

Term	Meaning
Active allocation	An allocation with status Provisional or Confirmed whose end date has not passed.
Audit log	Append-only record of security-relevant and data-changing actions (SEC-REQ-8).
Auto-Allocate	System function that allocates pending applications in priority order (FR-4.7).
Check-out	The act of a resident vacating a bed, ending the allocation.
Confirmed allocation	An allocation whose fee has been fully paid.
Gender designation	Attribute of a hostel stating whether it houses male or female students.
Master list	CSV file of enrolled students provided by the university records office.
Occupancy	Number of active allocations in a room; occupancy % = occupied beds / total beds x 100.
Provisional allocation	An allocation awaiting full fee payment (BR-5).
Soft delete	Marking a record inactive instead of removing it from the database.
Under Maintenance	Room status in which no beds can be allocated.

Appendix B: Field Layouts

Tables B-1 to B-5 give the field layouts and properties of the main data-entry forms. Table B-6 lists the report contents.

Table B-1: Student registration and profile

Field	Length	Data Type	Description	Is Mandatory
SRN	13	Alphanumeric	University registration number, from master list (read-only)	Y
Student Name	60	String	Full name, from master list (read-only)	Y
Gender	1	Enum (M/F)	From master list (read-only)	Y
Programme / Year	30 / 1	String / Numeric	e.g. B.Tech CSE / 1-4 (read-only)	Y
Institutional E-mail	100	E-mail	Login ID; must match master list	Y
Mobile Number	10	Numeric	Starts with 6-9	Y
Permanent Address	250	String	10-250 characters	Y
Guardian Name / Relationship	60 / 20	String	Parent or legal guardian	Y
Guardian Mobile	10	Numeric	Starts with 6-9	Y
Emergency Contact Name / Number	60 / 10	String / Numeric	If different from guardian	N
Blood Group	3	Enum	A+, A-, B+, B-, AB+, AB-, O+, O-	Y
Photograph	2 MB	Image (JPEG/PNG)	Passport-style photo	N

Table B-2: Room

Field	Length	Data Type	Description	Is Mandatory
Hostel	-	Foreign key	Hostel containing the room	Y
Room Number	10	Alphanumeric	Unique within hostel	Y
Floor	2	Numeric	0-20	Y
Room Type	10	Enum	Single / Double / Triple	Y
Capacity	1	Numeric	1-4 beds	Y
Annual Fee	9,2	Decimal	INR 0.00 - 5,00,000.00	Y
Status	20	Enum	Available / Full / Under Maintenance	Y

Table B-3: Room application

Field	Length	Data Type	Description	Is Mandatory
Application Number	15	Alphanumeric	System generated, APP-YYYY-NNNNN	Y
Academic Year	9	String	e.g. 2026-2027	Y
Preference 1 / 2 / 3	10	Enum	Room type; distinct values	Y / N / N
Preferred Roommate SRN	13	Alphanumeric	Must exist in master list	N
Status	20	Enum	Pending / Allocated / Rejected / Withdrawn	Y
Rejection Reason	300	String	Mandatory when status is Rejected	N
Submitted At	-	Date-time	System generated	Y

Table B-4: Complaint

Field	Length	Data Type	Description	Is Mandatory
Complaint ID	14	Alphanumeric	CMP-YYYY-NNNNN, system generated	Y
Category	15	Enum	Electrical, Plumbing, Furniture, Cleanliness, Internet, Other	Y
Description	500	String	20-500 characters	Y
Photo	2 MB	Image	JPEG/PNG	N
Status / Remark	15 / 300	Enum / String	Open, In Progress, Resolved, Rejected / warden remark	Y

Table B-5: Fee payment

Field	Length	Data Type	Description	Is Mandatory
Receipt Number	15	Alphanumeric	RCP-YYYY-NNNNN, sequential, system generated	Y
Allocation	-	Foreign key	Allocation the payment is for	Y
Amount	9,2	Decimal	Greater than 0 and not above balance	Y
Payment Mode	15	Enum	Online transfer / Demand draft / Cash	Y
Transaction Reference	30	Alphanumeric	Bank or DD reference	Y (except Cash)
Payment Date	8	Date	Not in the future	Y

Table B-6: Report requirements (fields included in each report)

Occupancy Report	Resident List	Fee Dues Report	Complaint Summary
Hostel	SRN	SRN	Complaint ID
Floor	Student Name	Student Name	Hostel / Room
Room Number	Hostel / Room / Bed	Hostel / Room	Category
Room Type	Year of Study	Annual Fee	Raised On
Capacity	Mobile Number	Amount Paid	Status
Occupied Beds	Guardian Mobile	Balance	Days Open
Vacant Beds	Blood Group	Allocation Status	Resolved On
Status	Allocation Start Date	Days Since Allocation	Warden Remark
Occupancy %			

Appendix C: Requirement Traceability Matrix

Architecture, design, code and test references are the planned identifiers used in the Software Architecture and Design Specification and the Test Plan of this project; they will be updated as those documents and the code evolve.

Sl. No	Requirement ID	Brief Description of Requirement	Architecture Reference	Design Reference	Code File Reference	Test Case ID	System Test Case ID
1	FR-1.1, 1.2	Student self-registration with verification	C1 Auth	SD-3 Register	accounts/views.py	TC-01	STC-01
2	FR-1.3, 1.5	Login and account lock-out	C1 Auth	SD-1 Login	accounts/views.py	TC-02, TC-03	STC-01
3	FR-1.4	Role-based access to pages	C1 Auth	RBAC decorators	core/permissions.py	TC-04	STC-08
4	FR-2.1 - 2.3	Edit and validate profile	C2 Profile	Profile form design	students/forms.py	TC-05	STC-02
5	FR-2.5, 2.6	Student search, hostel-scoped	C2 Profile	Search API	students/views.py	TC-06	STC-02
6	FR-3.2 - 3.4	Room definition and occupancy status	C3 Rooms	Room model	rooms/models.py	TC-07	STC-03
7	FR-4.2, 4.3	Submit one application in window	C4 Allocation	SD-2 Apply & Allocate	allocation/views.py	TC-08	STC-04
8	FR-4.5, 4.6	Gender-matched, no double allocation	C4 Allocation	SD-2 Apply & Allocate	allocation/services.py	TC-09, TC-10	STC-04
9	FR-4.7	Auto-allocate by priority	C4 Allocation	Auto-allocation algorithm	allocation/services.py	TC-11	STC-04
10	FR-5.2	Atomic room change	C4 Allocation	SD-4 Room change	allocation/services.py	TC-12	STC-05
11	FR-6.1 - 6.3	Raise and resolve complaint	C5 Complaints	Complaint state machine	complaints/views.py	TC-13	STC-06
12	FR-7.2 - 7.4	Record payment, receipt, confirm	C6 Fees	Payment API	fees/services.py	TC-14	STC-07
13	FR-8.2, 8.3	Reports and export	C7 Reports	Report service	reports/views.py	TC-15	STC-07
14	FR-9.1, 9.2	E-mail notifications with retry	C8 Notifications	Notification service	notifications/tasks.py	TC-16	STC-04
15	NFR-PERF-1	2 s page response at 200 users	Deployment view	-	-	TC-17	STC-09
16	NFR-SAFE-1	Daily backup and restore	Deployment view	-	ops/backup.sh	TC-18	STC-10

Sl. No	Requirement ID	Brief Description of Requirement	Architecture Reference	Design Reference	Code File Reference	Test Case ID	System Test Case ID
17	SEC-REQ-1, 2	Password hashing and policy	Security arch.	Auth settings	hms/settings.py	TC-19	STC-08
18	SEC-REQ-4	Server-side authorisation, no IDOR	Security arch.	RBAC decorators	core/permissions.py	TC-20	STC-08
19	SEC-REQ-6	CSRF/XSS/SQLi protection	Security arch.	Middleware config	hms/settings.py	TC-21	STC-08
20	SEC-REQ-8	Audit logging	Security arch.	Audit service	core/audit.py	TC-22	STC-08
21	NFR-USE-1	Application within 5 minutes	C4 Allocation (UI)	UI design	templates/allocation/	TC-23	STC-11

Index

Term	Section	Term	Section
Allocation	4.4, 5.5	Notifications	4.9
Analysis models	3.5	Performance	5.1
Audit log	2.2, 5.3.2	Permission matrix (BR-1)	5.5
Business rules	5.5	Product functions	2.2
Check-out	4.5	Reports	4.8, App. B
Complaints	4.6	Room change	4.5
Constraints	2.5	Safety	5.2
ER diagram	3.5.2	Security objectives	5.3.1
Fee tracking	4.7	Security requirements	5.3.2
Field layouts	App. B	Traceability matrix	App. C
Interfaces	3.1 - 3.4	Use case diagram	3.5.1
Login / authentication	4.1	User classes	2.3